PROJECT

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SET UP A RAG CHATBOT IN BEDROCK

AI-ML SERVICE

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Objective

This project aims to establish a scalable RAG chatbot using AWS Bedrock for accurate, contextually relevant responses. It will streamline document ingestion to user interaction via a centralized knowledge base.

The setup will optimize retrieval through S3 document storage and OpenSearch indexing within the knowledge base. This will enhance information accessibility and user satisfaction with Bedrock's AI models.

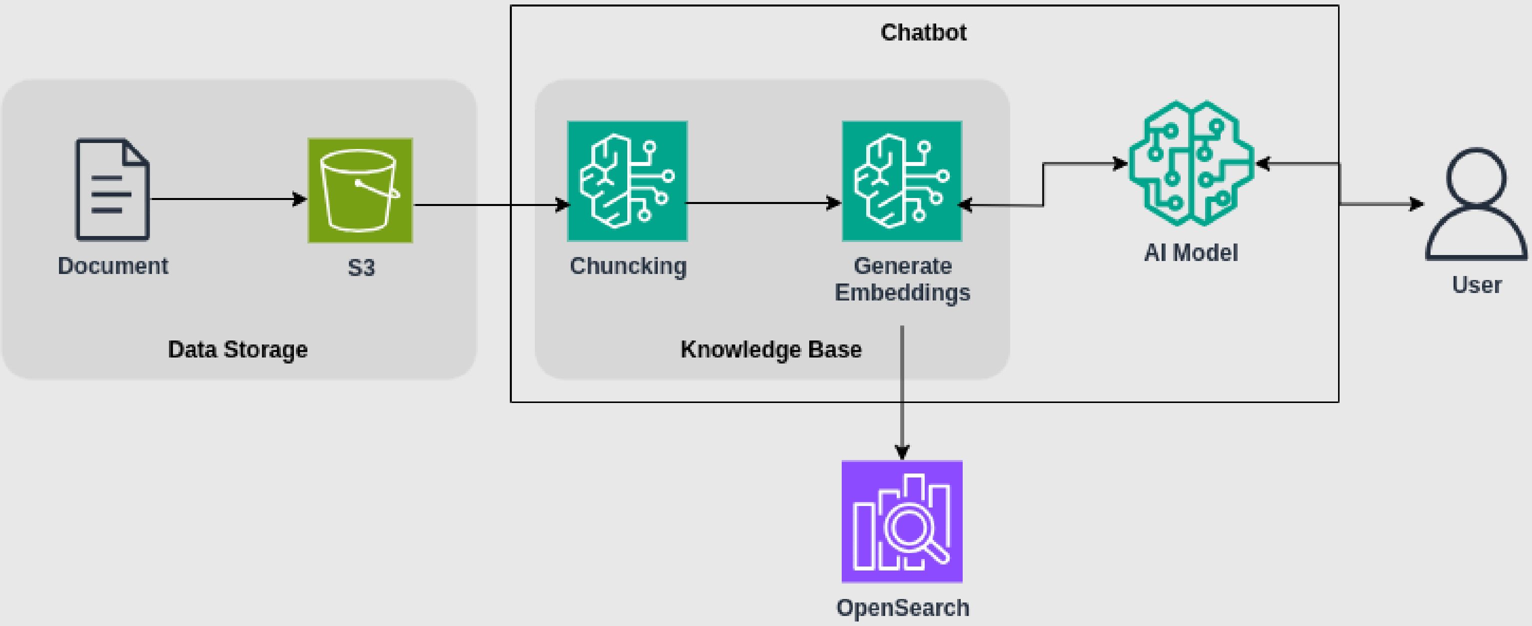
Services

Amazon Bedrock - Provides access to foundation models (FMs) from Amazon and leading AI companies via an API.

Amazon S3 - Offers secure, durable, and highly scalable object storage for various data, including documents.

Amazon OpenSearch Service - A managed service that simplifies the deployment, operation, and scaling of OpenSearch clusters for search and analytics.

Architecture Diagram



Documents are uploaded to Amazon S3, then chunked and embedded using Bedrock. These embeddings are stored in OpenSearch to create a Knowledge Base. When a user asks a question, relevant chunks are retrieved and sent to an AI model in Bedrock, which generates a smart response for the user.

Steps to Implement

1. Store Documents in Amazon S3:
Open the S3 console and create a new bucket with a unique name. Upload your documents (PDFs, text, etc.) that contain the knowledge your chatbot will use. Make sure the bucket permissions allow access for Bedrock to read them.

2. Set Up a Knowledge Base in Bedrock:
Go to the Amazon Bedrock console and navigate to "Knowledge Base." Click on "Create Knowledge Base," give it a name, and select the S3 bucket containing your documents. Enable automatic chunking so Bedrock can split the content meaningfully.

3. Connect to OpenSearch for Embeddings:
In the same setup, choose OpenSearch as the vector store. Bedrock will generate embeddings for each chunk and index them into OpenSearch, making them ready for efficient retrieval during chats.

4. Enable AI Model Access and Sync:
Select a foundation model like Anthropic Claude or Amazon Titan for generating responses. After model selection, sync your Knowledge Base to begin embedding generation and linking to the chatbot logic.

5. Chat with Your RAG Chatbot:
Once syncing is complete, go to the chat playground in Bedrock. Ask questions related to your documents. The chatbot will retrieve relevant content from OpenSearch and use the AI model to generate answers in real time.

Story Time

Amazon Bedrock (The AI Spellbook):
Think of Amazon Bedrock as your ultimate spellbook filled with magical AI powers. You whisper a question into it, and it summons the perfect generative response from its enchanted pages, guided by your knowledge scrolls.

Amazon S3 (The Document Vault):
Amazon S3 is your sturdy vault where all your precious documents are stored safely. Whether it's tales, guides, or secret recipes, S3 keeps them ready for your chatbot to learn from.

Amazon OpenSearch (The Memory Library):
OpenSearch acts as your magical memory library. It takes all the knowledge scrolls, turns them into glowing runes (embeddings), and arranges them neatly on shelves so the AI can instantly find the right spell (info) when asked.

BY
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THANK YOU!

AI/ML PRO
IN THE MAKING